Factor Analysis

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.885
Bartlett's Test of Sphericity	Approx. Chi-Square	64925.704
	df	496
	Sig.	.000

Communalities

	Initial	Extraction
TEL1	.578	.644
TEL2	.632	.718
TEL3	.669	.772
TEL4	.580	.646
EL1	.594	.667
EL2	.632	.709
EL3	.675	.774
EL4	.586	.654
WIL1	.582	.651
WIL2	.618	.697
WIL3	.681	.783
WIL4	.595	.668
DCC1	.597	.661
DCC2	.630	.706
DCC3	.692	.796
DCC4	.600	.667
PCSE1	.584	.653
PCSE2	.630	.713
PCSE3	.675	.772
PCSE4	.606	.678
CA1	.602	.674
CA2	.643	.727
CA3	.670	.762
CA4	.585	.655
ICS1	.592	.667
ICS2	.621	.701
ICS3	.673	.776

Communalities

	Initial	Extraction
ICS4	.583	.653
GPE1	.605	.674
GPE2	.647	.725
GPE3	.681	.776
GPE4	.583	.647

Extraction Method: Principal Axis Factoring.

Total Variance Explained

	Initial Eigenvalues			Extraction Sums of Squared Loadings	
Factor	Total	% of Variance	Cumulative %	Total	% of Variance
1	6.118	19.120	19.120	5.824	18.201
2	4.215	13.171	32.291	3.919	12.248
3	3.607	11.272	43.563	3.311	10.347
4	3.110	9.719	53.282	2.812	8.788
5	2.378	7.433	60.715	2.083	6.509
6	2.015	6.296	67.010	1.718	5.369
7	1.779	5.560	72.571	1.482	4.633
8	1.608	5.026	77.597	1.316	4.111
9	.378	1.182	78.778		
10	.366	1.145	79.923		
11	.360	1.125	81.048		
12	.354	1.107	82.156		
13	.351	1.098	83.253		
14	.339	1.059	84.312		
15	.336	1.051	85.363		
16	.330	1.031	86.394		
17	.323	1.011	87.405		
18	.320	.999	88.404		
19	.310	.970	89.374		
20	.306	.958	90.331		
21	.300	.937	91.269		
22	.297	.929	92.198		
23	.292	.912	93.109		

Total Variance Explained

	Extraction Sums of Squared ...	Rotation Sums of Squared Loadings[a]
Factor	Cumulative %	Total
1	18.201	3.707
2	30.449	3.515
3	40.796	3.422
4	49.584	3.443
5	56.093	3.328
6	61.462	2.839
7	66.095	3.591
8	70.206	4.376
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		

Total Variance Explained

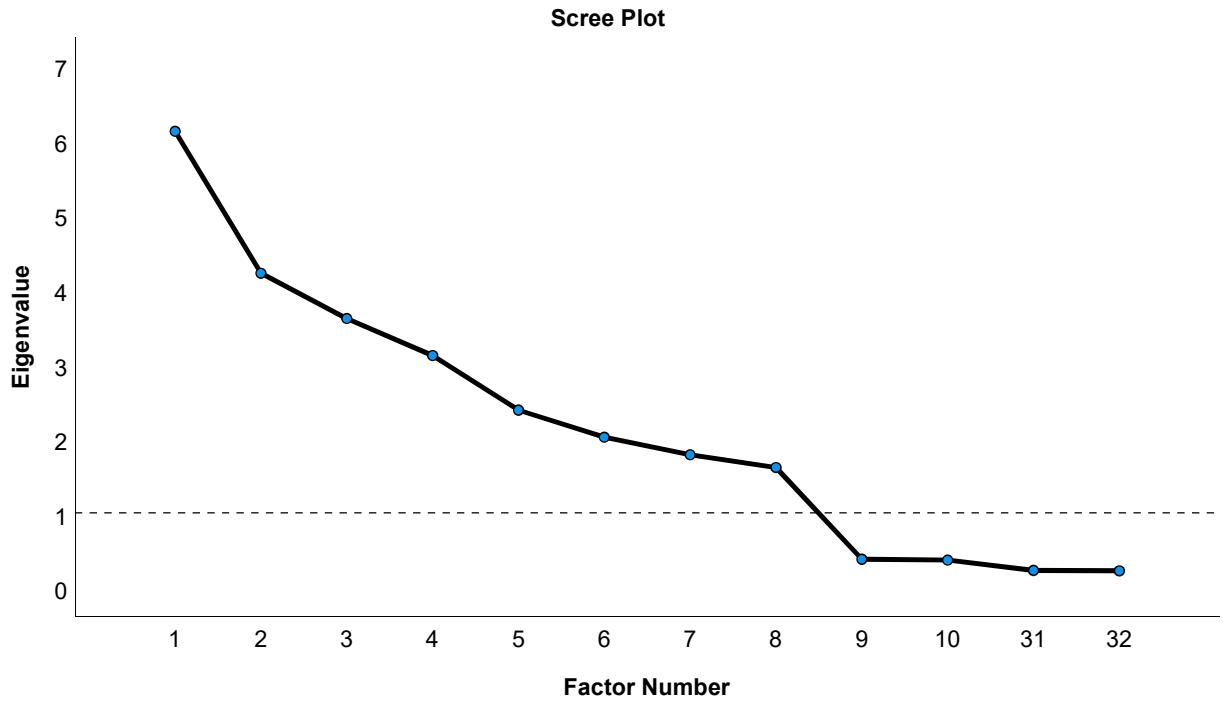
Factor	Initial Eigenvalues			Extraction Sums of Squared Loadings	
	Total	% of Variance	Cumulative %	Total	% of Variance
24	.282	.882	93.991		
25	.260	.811	94.803		
26	.253	.792	95.595		
27	.248	.776	96.370		
28	.240	.750	97.120		
29	.239	.745	97.866		
30	.233	.728	98.593		
31	.227	.711	99.304		
32	.223	.696	100.000		

Total Variance Explained

Factor	Extraction Sums of Squared ...	Rotation Sums of Squared Loadings[a]
	Cumulative %	Total
24		
25		
26		
27		
28		
29		
30		
31		
32		

Extraction Method: Principal Axis Factoring.

a. When factors are correlated, sums of squared loadings cannot be added to obtain a total variance.



Factor Matrix[a]

	Factor					
	1	2	3	4	5	6
TEL1						
TEL2						
TEL3						
TEL4						
EL1			-.536			
EL2			-.560			
EL3			-.589			
EL4			-.532			
WIL1		.576				
WIL2		.598				
WIL3		.638				
WIL4		.596				
DCC1						
DCC2						
DCC3	.515					
DCC4						

Factor Matrix[a]

	Factor	
	7	8
TEL1		
TEL2		
TEL3		
TEL4		
EL1		
EL2		
EL3		
EL4		
WIL1		
WIL2		
WIL3		
WIL4		
DCC1		
DCC2		
DCC3		
DCC4		

Factor Matrix[a]

	Factor					
	1	2	3	4	5	6
PCSE1						
PCSE2						
PCSE3						
PCSE4						
CA1		.561				
CA2		.585				
CA3		.605				
CA4		.553				
ICS1				.780		
ICS2				.806		
ICS3				.842		
ICS4				.775		
GPE1	.665					
GPE2	.704					
GPE3	.719					
GPE4	.659					

Factor Matrix[a]

	Factor	
	7	8
PCSE1		
PCSE2		
PCSE3		
PCSE4		
CA1		
CA2		
CA3		
CA4		
ICS1		
ICS2		
ICS3		
ICS4		
GPE1		
GPE2		
GPE3		
GPE4		

Extraction Method: Principal Axis Factoring.

a. 8 factors extracted. 6 iterations required.

Pattern Matrix[a]

	Factor					
	1	2	3	4	5	6
TEL1						
TEL2						
TEL3						
TEL4						
EL1				.816		
EL2				.842		
EL3				.878		
EL4				.808		
WIL1					.807	
WIL2					.834	
WIL3					.883	
WIL4					.816	
DCC1	.811					
DCC2	.840					
DCC3	.894					
DCC4	.810					
PCSE1		.807				
PCSE2		.851				
PCSE3		.876				
PCSE4		.818				
CA1			.818			
CA2			.856			
CA3			.867			
CA4			.810			
ICS1						.816
ICS2						.838
ICS3						.882
ICS4						.806
GPE1						
GPE2						
GPE3						
GPE4						

Pattern Matrix[a]

	Factor	
	7	8
TEL1	.795	
TEL2	.857	
TEL3	.879	
TEL4	.794	
EL1		
EL2		
EL3		
EL4		
WIL1		
WIL2		
WIL3		
WIL4		
DCC1		
DCC2		
DCC3		
DCC4		
PCSE1		
PCSE2		
PCSE3		
PCSE4		
CA1		
CA2		
CA3		
CA4		
ICS1		
ICS2		
ICS3		
ICS4		
GPE1		.826
GPE2		.840
GPE3		.882
GPE4		.800

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.[a]

a. Rotation converged in 6 iterations.

Structure Matrix

	Factor					
	1	2	3	4	5	6
TEL1						
TEL2						
TEL3						
TEL4						
EL1				.817		
EL2				.842		
EL3				.880		
EL4				.808		
WIL1					.806	
WIL2					.834	
WIL3					.885	
WIL4					.817	
DCC1	.812					
DCC2	.840					
DCC3	.892					
DCC4	.816					
PCSE1		.808				
PCSE2		.844				
PCSE3		.879				
PCSE4		.823				
CA1			.821			
CA2			.852			
CA3			.873			
CA4			.809			
ICS1						.816
ICS2						.837
ICS3						.881
ICS4						.808
GPE1						
GPE2						
GPE3						
GPE4						

Structure Matrix

	Factor	
	7	8
TEL1	.802	
TEL2	.847	
TEL3	.878	
TEL4	.803	
EL1		
EL2		
EL3		
EL4		
WIL1		
WIL2		
WIL3		
WIL4		
DCC1		
DCC2		
DCC3		
DCC4		
PCSE1		
PCSE2		
PCSE3		
PCSE4		
CA1		
CA2		
CA3		
CA4		
ICS1		
ICS2		
ICS3		
ICS4		
GPE1		.821
GPE2		.851
GPE3		.881
GPE4		.804

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.

Factor Correlation Matrix

Factor	1	2	3	4	5	6
1	1.000	.102	.026	.146	-.033	-.010
2	.102	1.000	.035	.348	-.002	.005
3	.026	.035	1.000	.119	.346	.000
4	.146	.348	.119	1.000	-.008	-.012
5	-.033	-.002	.346	-.008	1.000	.047
6	-.010	.005	.000	-.012	.047	1.000
7	.394	.139	-.027	-.001	-.008	.020
8	.350	.309	.278	.262	.252	.109

Factor Correlation Matrix

Factor	7	8
1	.394	.350
2	.139	.309
3	-.027	.278
4	-.001	.262
5	-.008	.252
6	.020	.109
7	1.000	.332
8	.332	1.000

Extraction Method: Principal Axis Factoring.
Rotation Method: Promax with Kaiser Normalization.

Multilayer Perceptron

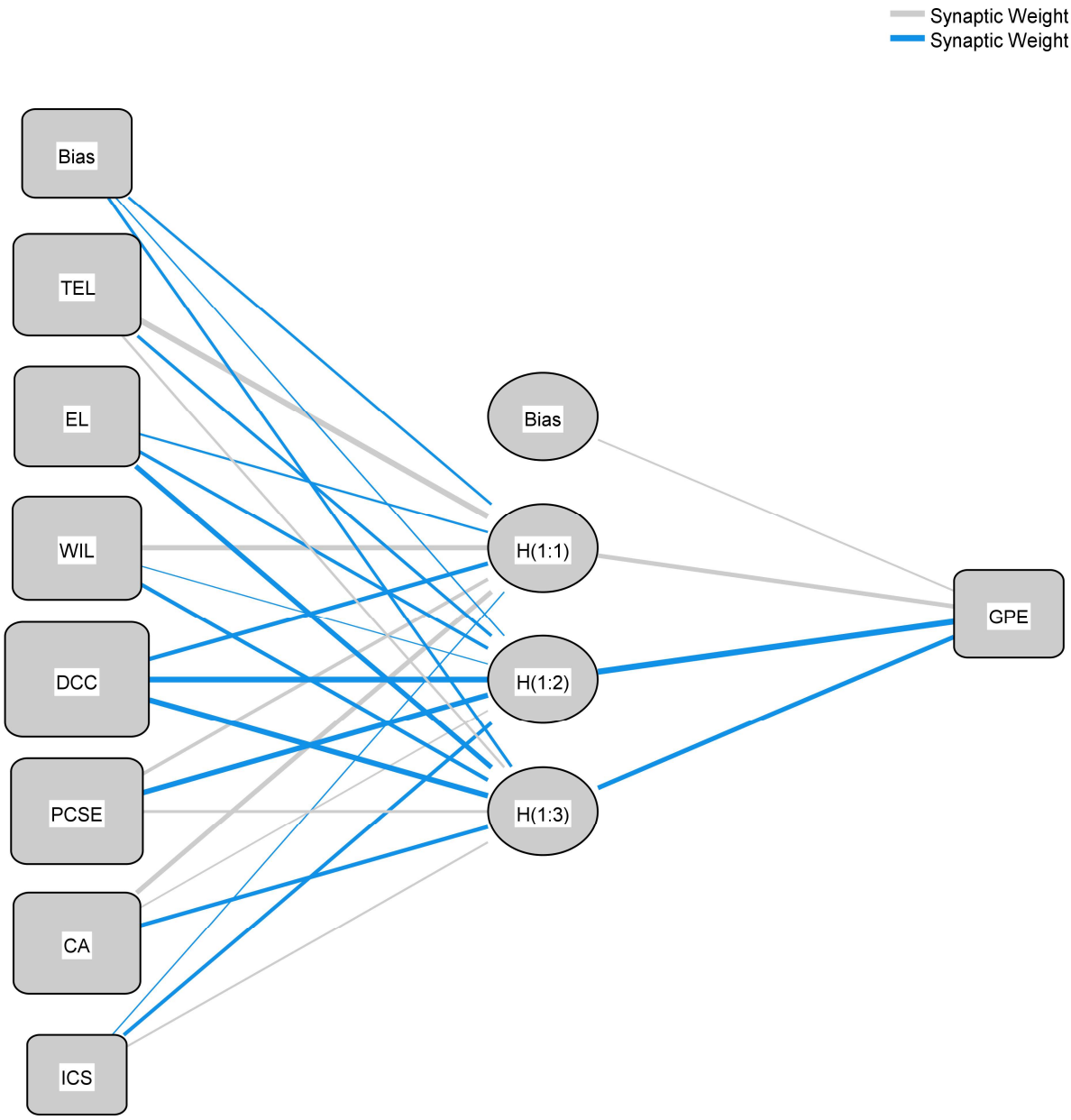
Case Processing Summary

		N	Percent
Sample	Training	2146	69.2%
	Testing	954	30.8%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

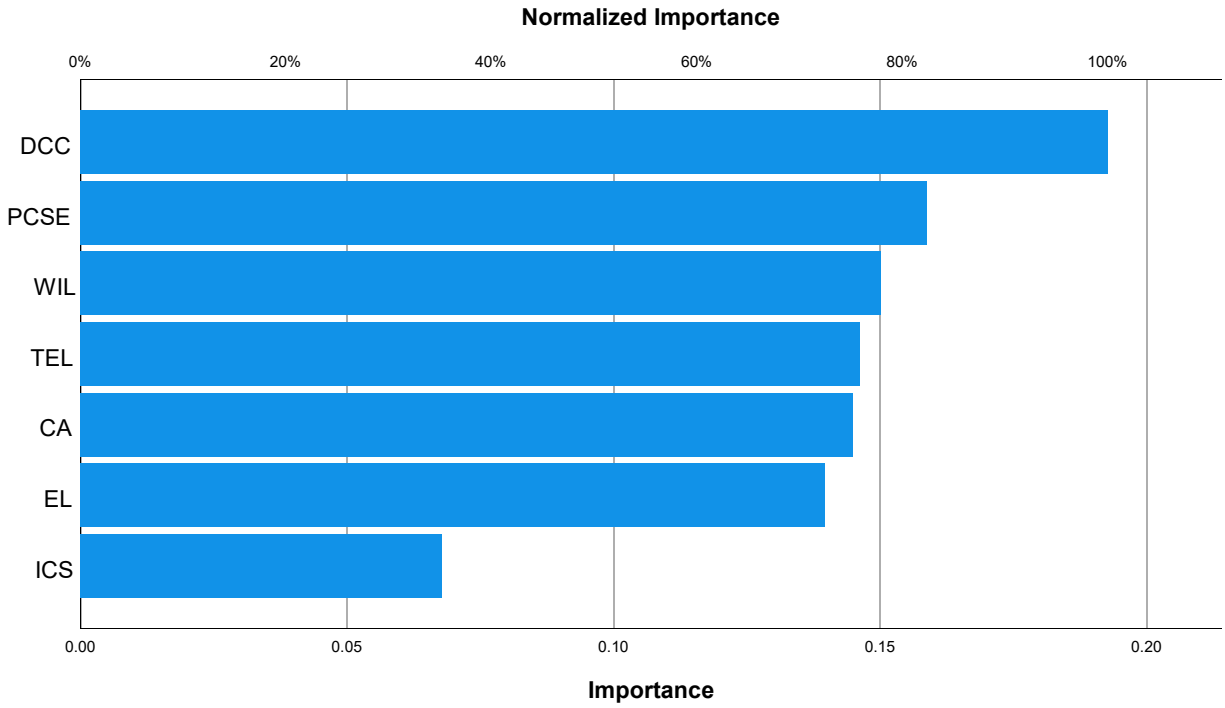
Training	Sum of Squares Error	754.868
	Relative Error	.704
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.10
Testing	Sum of Squares Error	317.592
	Relative Error	.692

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.146	75.8%
EL	.140	72.4%
WIL	.150	77.9%
DCC	.193	100.0%
PCSE	.159	82.4%
CA	.145	75.2%
ICS	.068	35.1%



Multilayer Perceptron

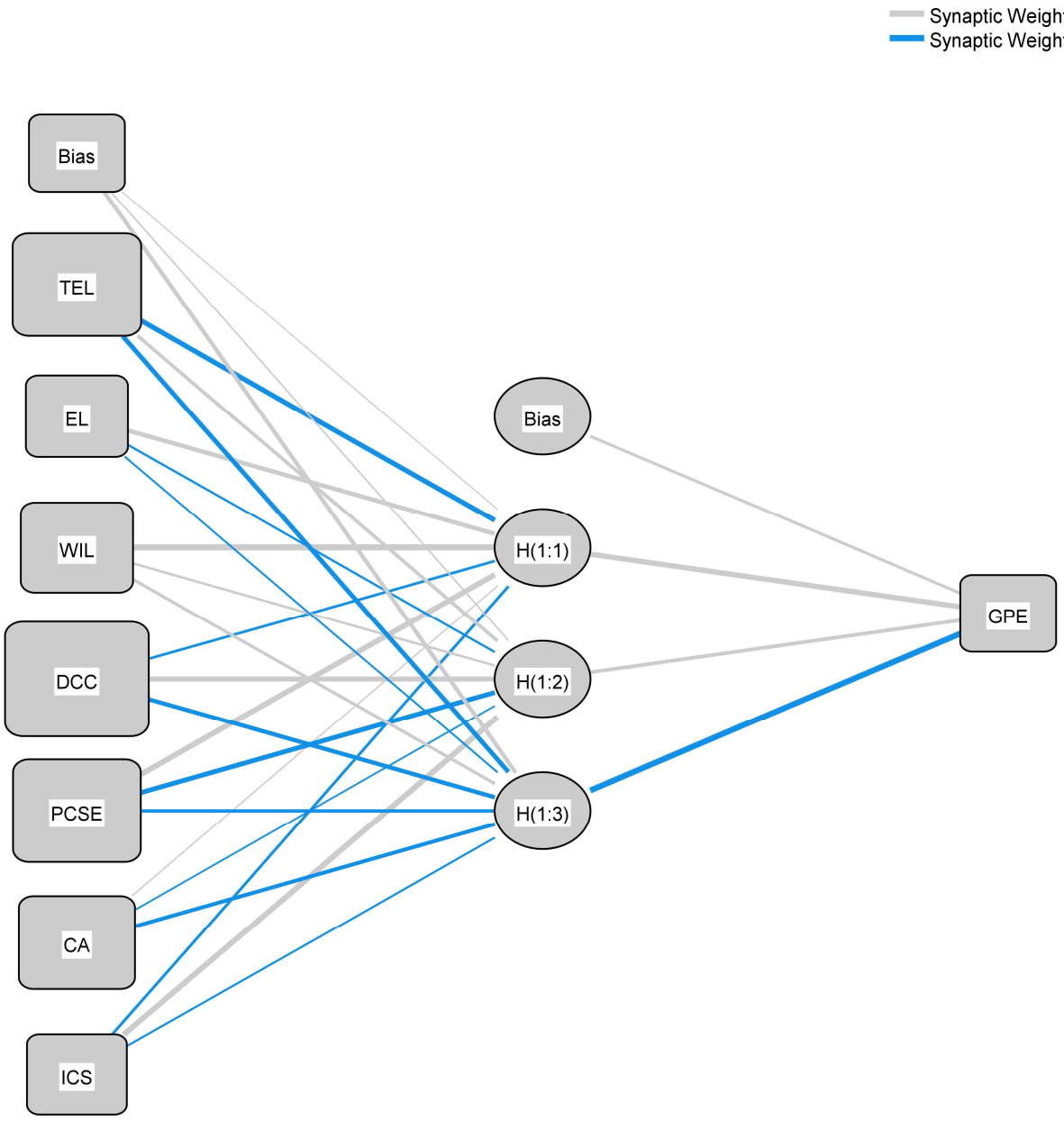
Case Processing Summary

		N	Percent
Sample	Training	2178	70.3%
	Testing	922	29.7%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

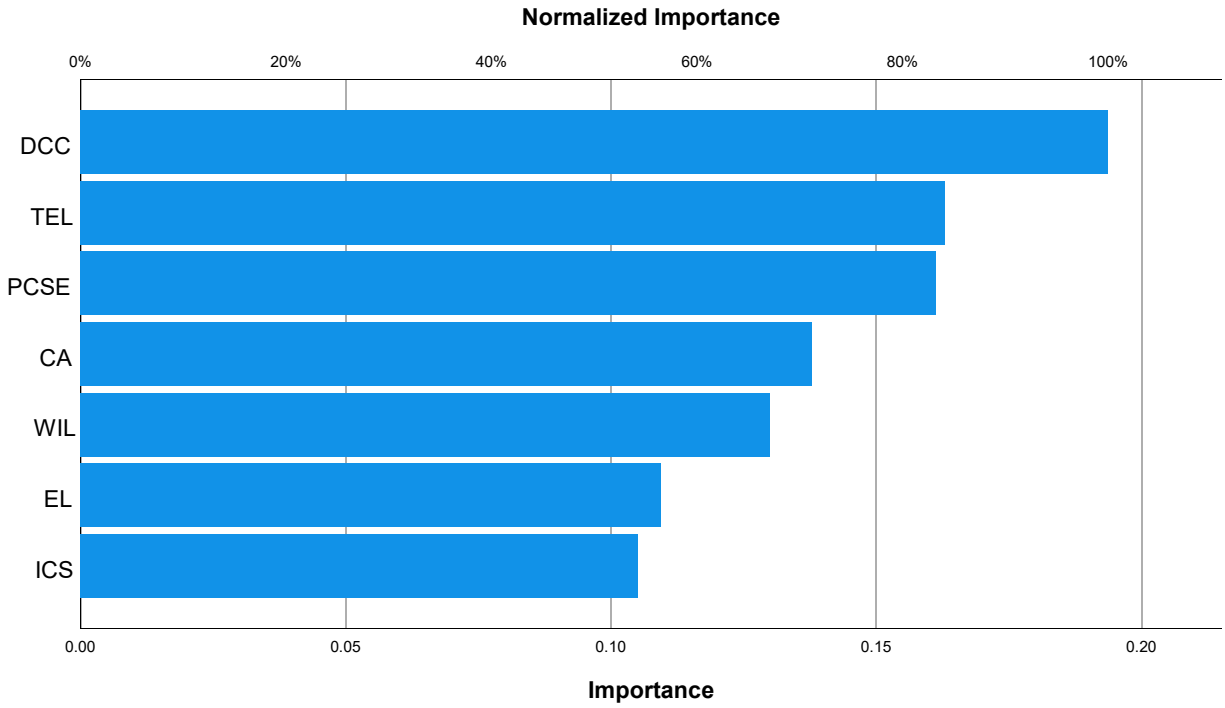
Training	Sum of Squares Error	756.876
	Relative Error	.695
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.13
Testing	Sum of Squares Error	323.720
	Relative Error	.710

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.163	84.1%
EL	.109	56.5%
WIL	.130	67.1%
DCC	.194	100.0%
PCSE	.161	83.3%
CA	.138	71.1%
ICS	.105	54.2%



Multilayer Perceptron

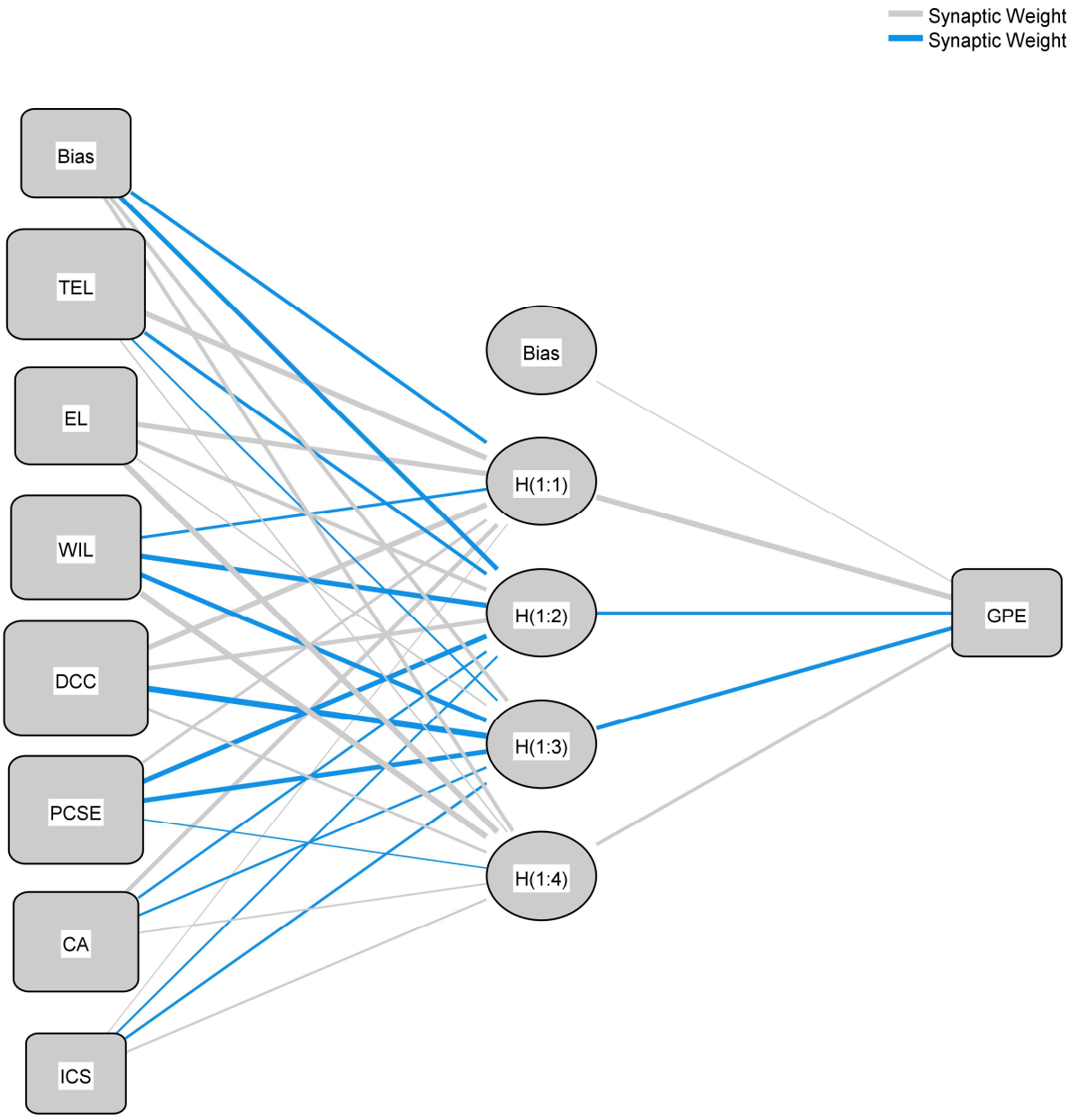
Case Processing Summary

		N	Percent
Sample	Training	2134	68.8%
	Testing	966	31.2%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

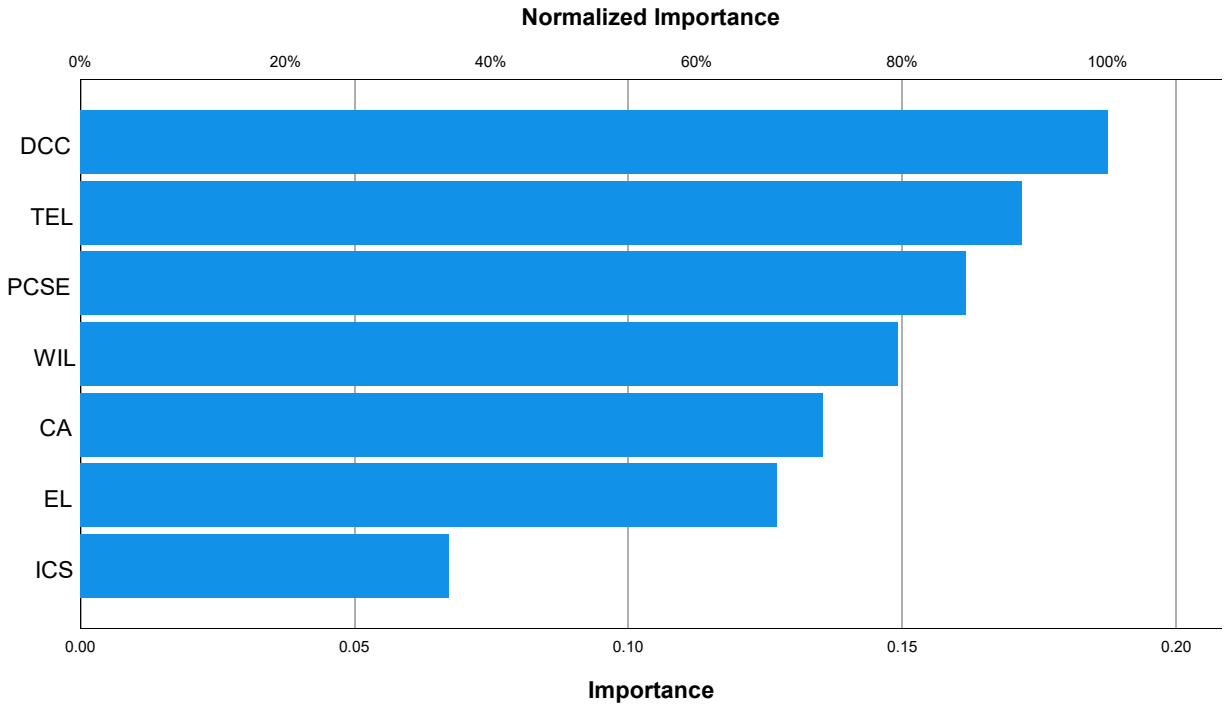
Training	Sum of Squares Error	736.296
	Relative Error	.690
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.10
Testing	Sum of Squares Error	336.283
	Relative Error	.710

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.172	91.6%
EL	.127	67.8%
WIL	.149	79.5%
DCC	.188	100.0%
PCSE	.162	86.2%
CA	.136	72.3%
ICS	.067	35.8%



Multilayer Perceptron

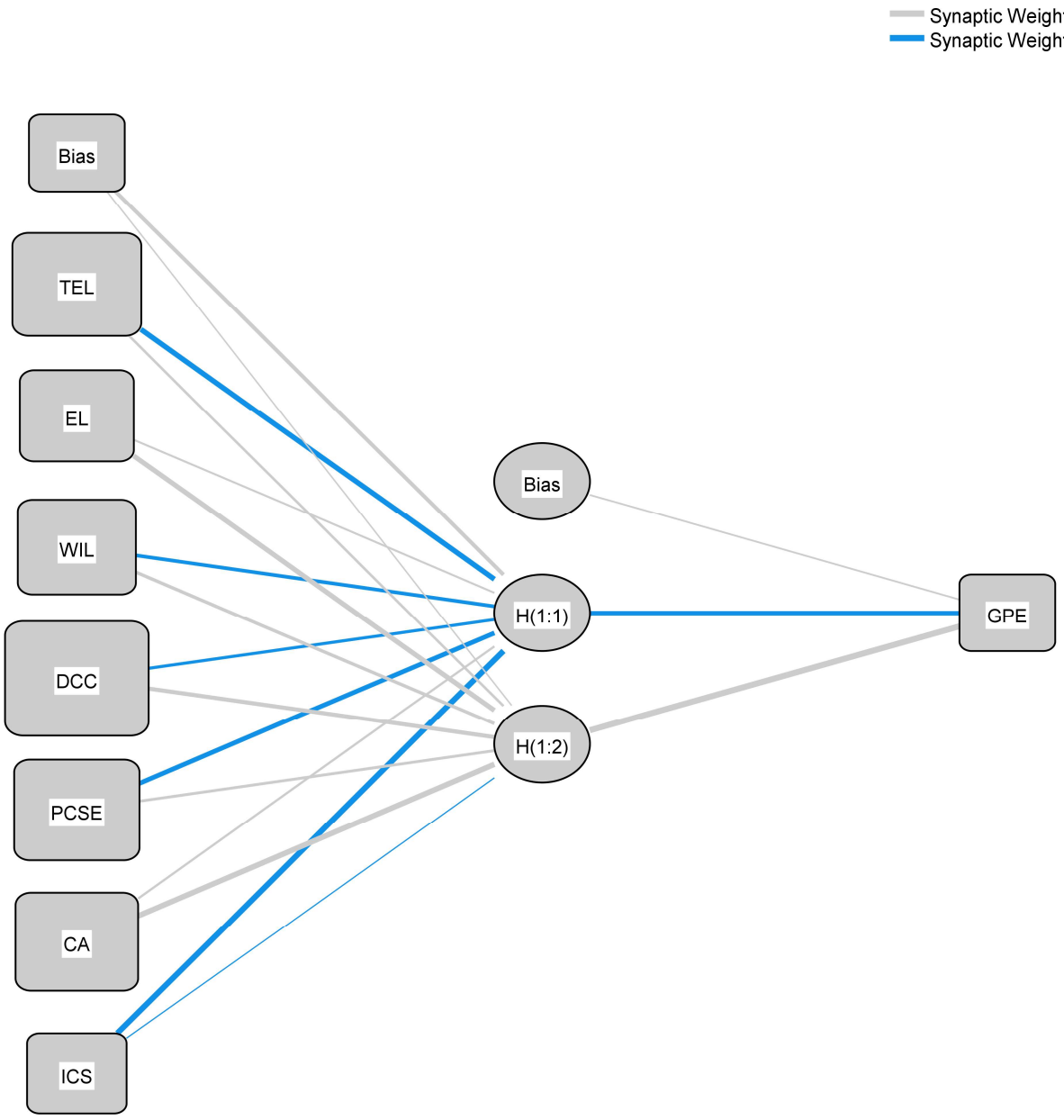
Case Processing Summary

		N	Percent
Sample	Training	2185	70.5%
	Testing	915	29.5%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

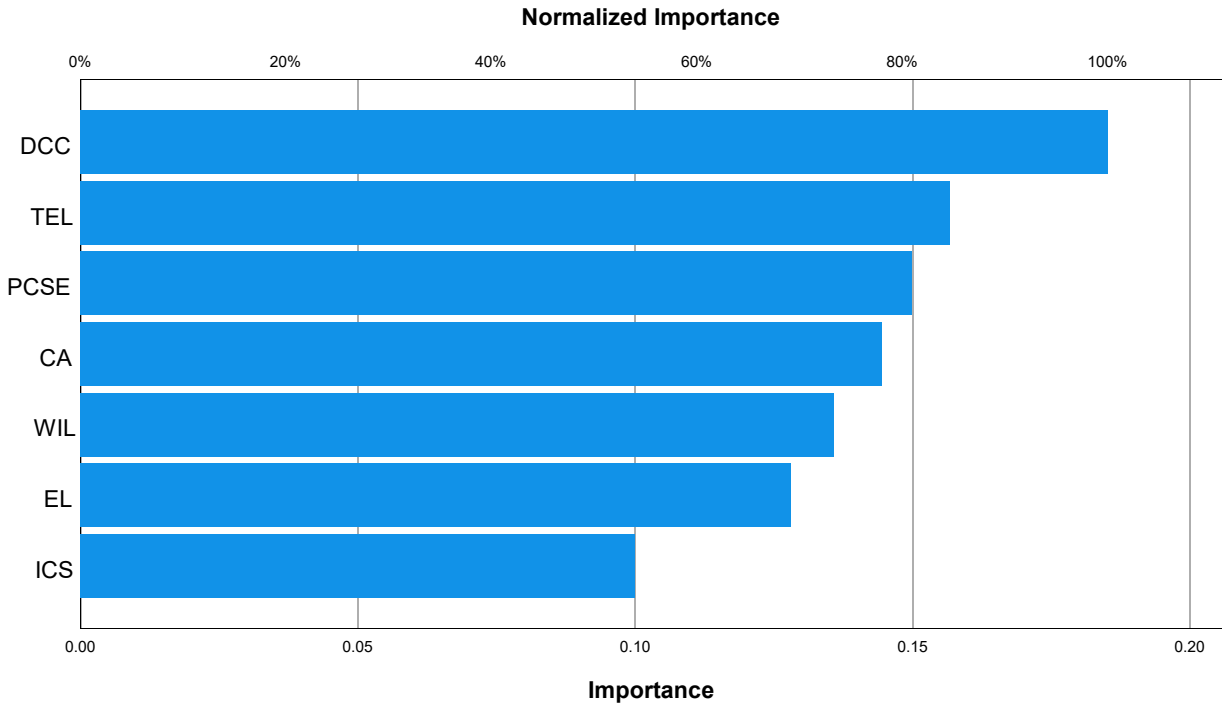
Training	Sum of Squares Error	760.427
	Relative Error	.696
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.13
Testing	Sum of Squares Error	324.972
	Relative Error	.707

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.157	84.6%
EL	.128	69.1%
WIL	.136	73.4%
DCC	.185	100.0%
PCSE	.150	80.9%
CA	.144	78.0%
ICS	.100	53.9%



Multilayer Perceptron

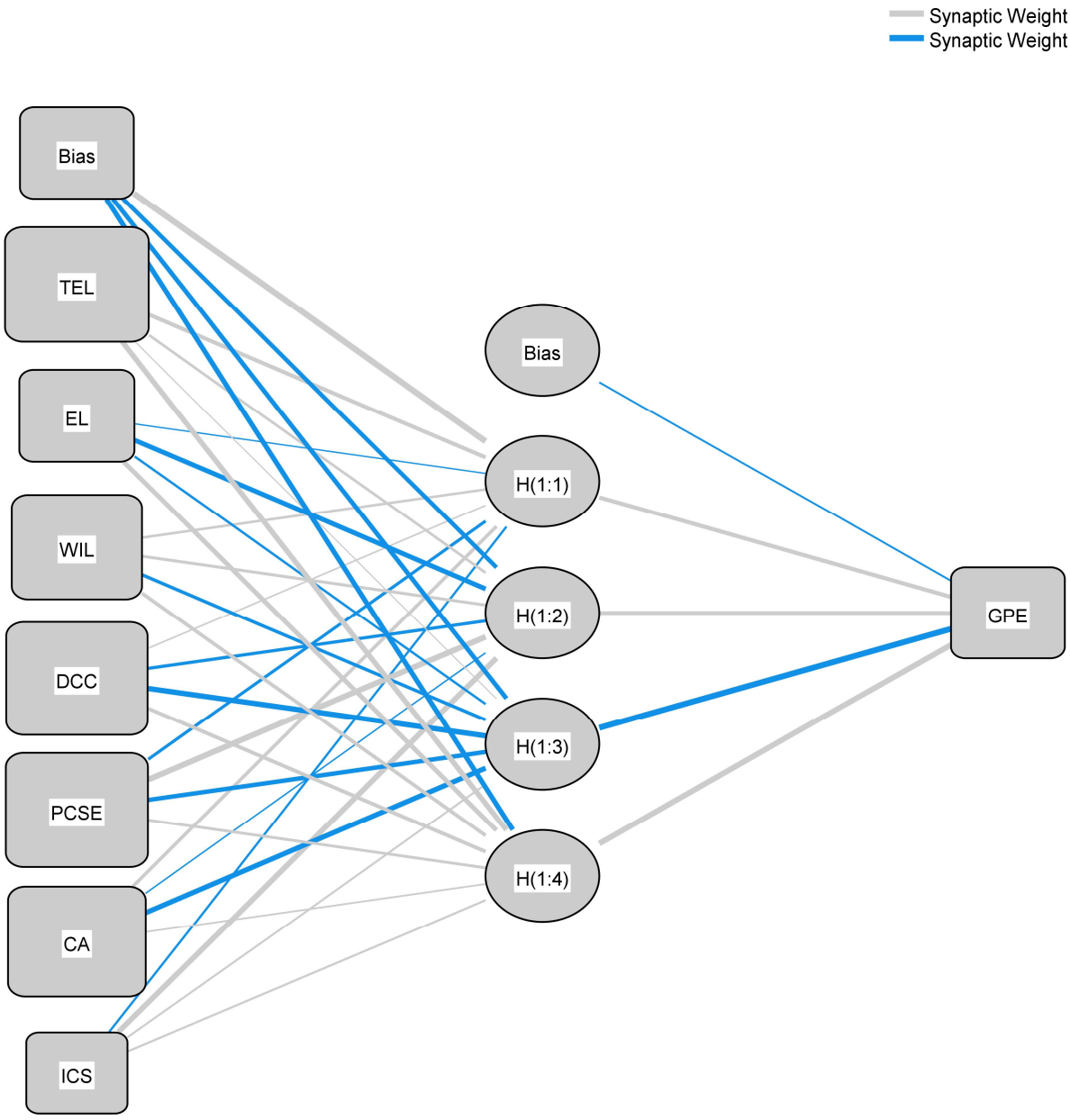
Case Processing Summary

		N	Percent
Sample	Training	2192	70.7%
	Testing	908	29.3%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

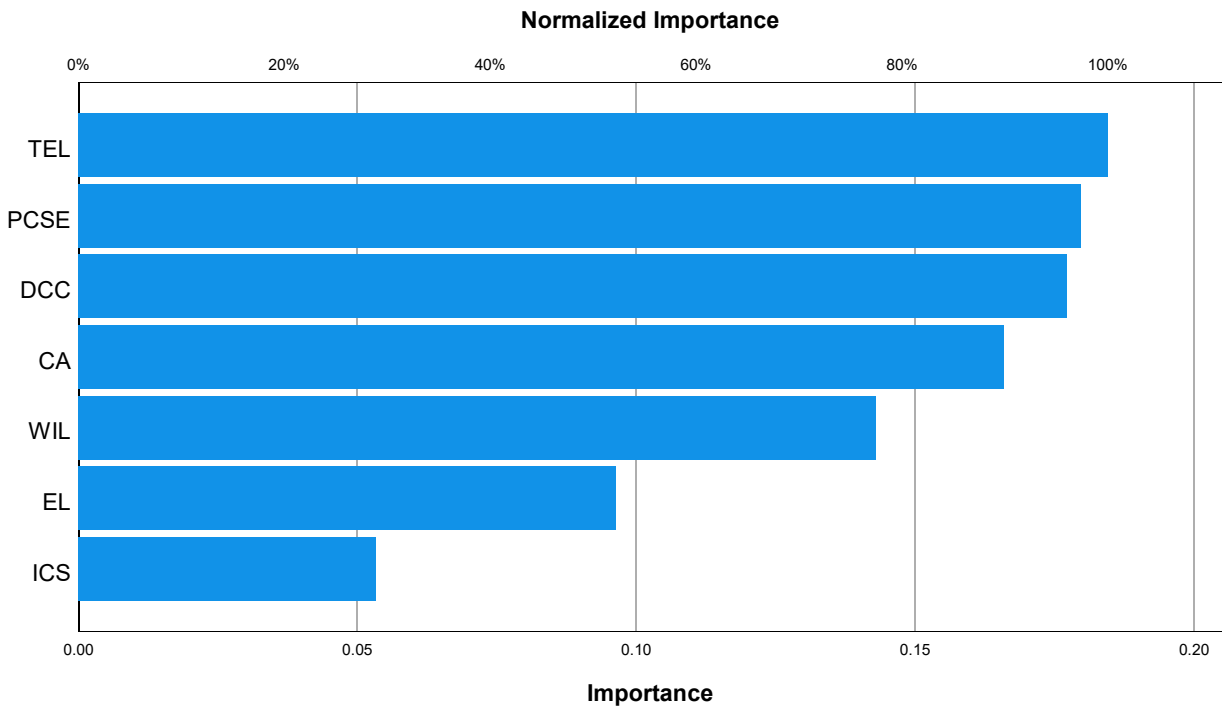
Training	Sum of Squares Error	751.753
	Relative Error	.686
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.09
Testing	Sum of Squares Error	321.203
	Relative Error	.717

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.185	100.0%
EL	.096	52.2%
WIL	.143	77.5%
DCC	.177	96.1%
PCSE	.180	97.4%
CA	.166	89.9%
ICS	.053	28.9%



Multilayer Perceptron

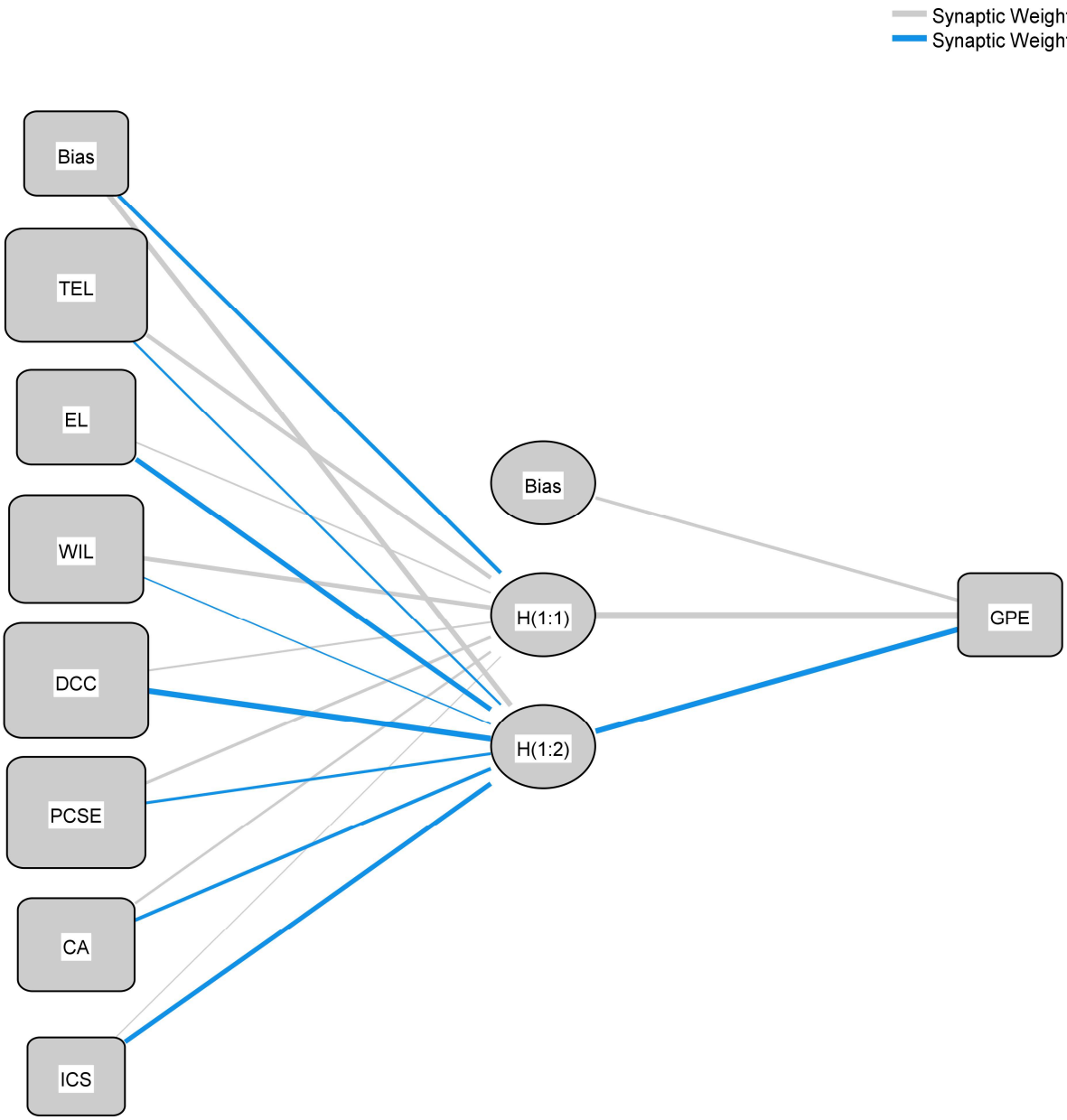
Case Processing Summary

		N	Percent
Sample	Training	2168	69.9%
	Testing	932	30.1%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

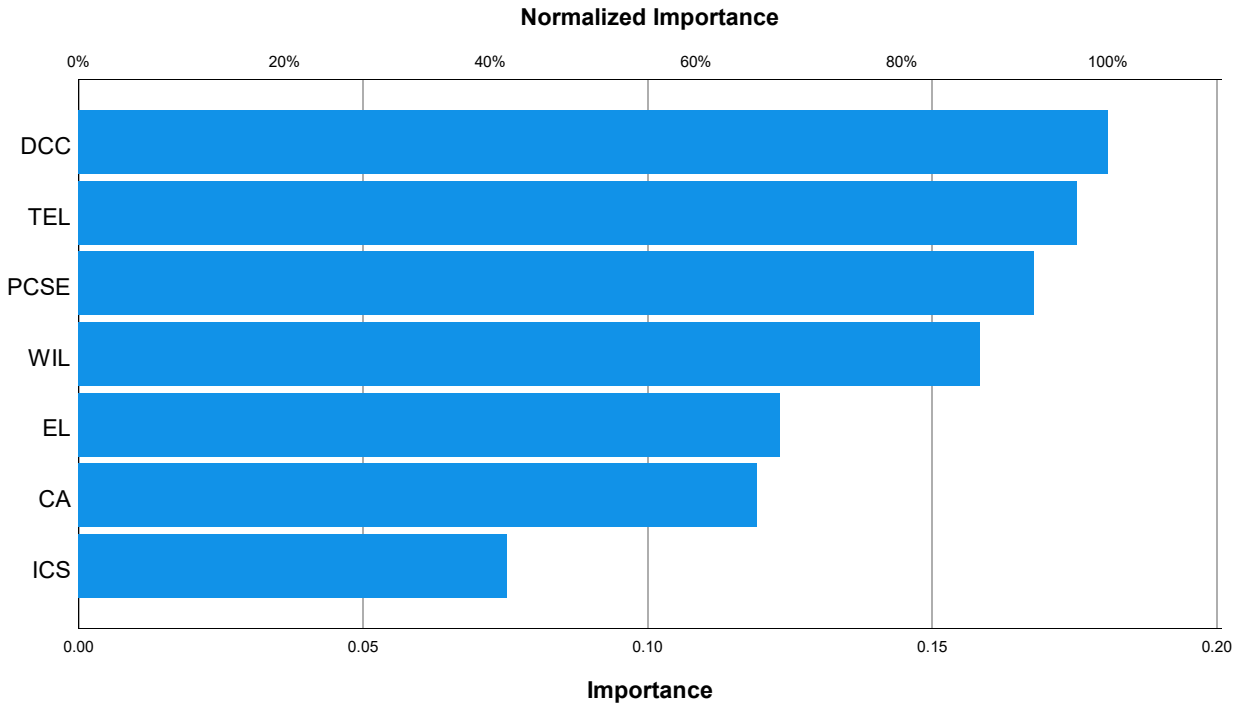
Training	Sum of Squares Error	755.406
	Relative Error	.697
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.11
Testing	Sum of Squares Error	341.070
	Relative Error	.702

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.175	97.0%
EL	.123	68.1%
WIL	.158	87.5%
DCC	.181	100.0%
PCSE	.168	92.8%
CA	.119	65.9%
ICS	.075	41.6%



Multilayer Perceptron

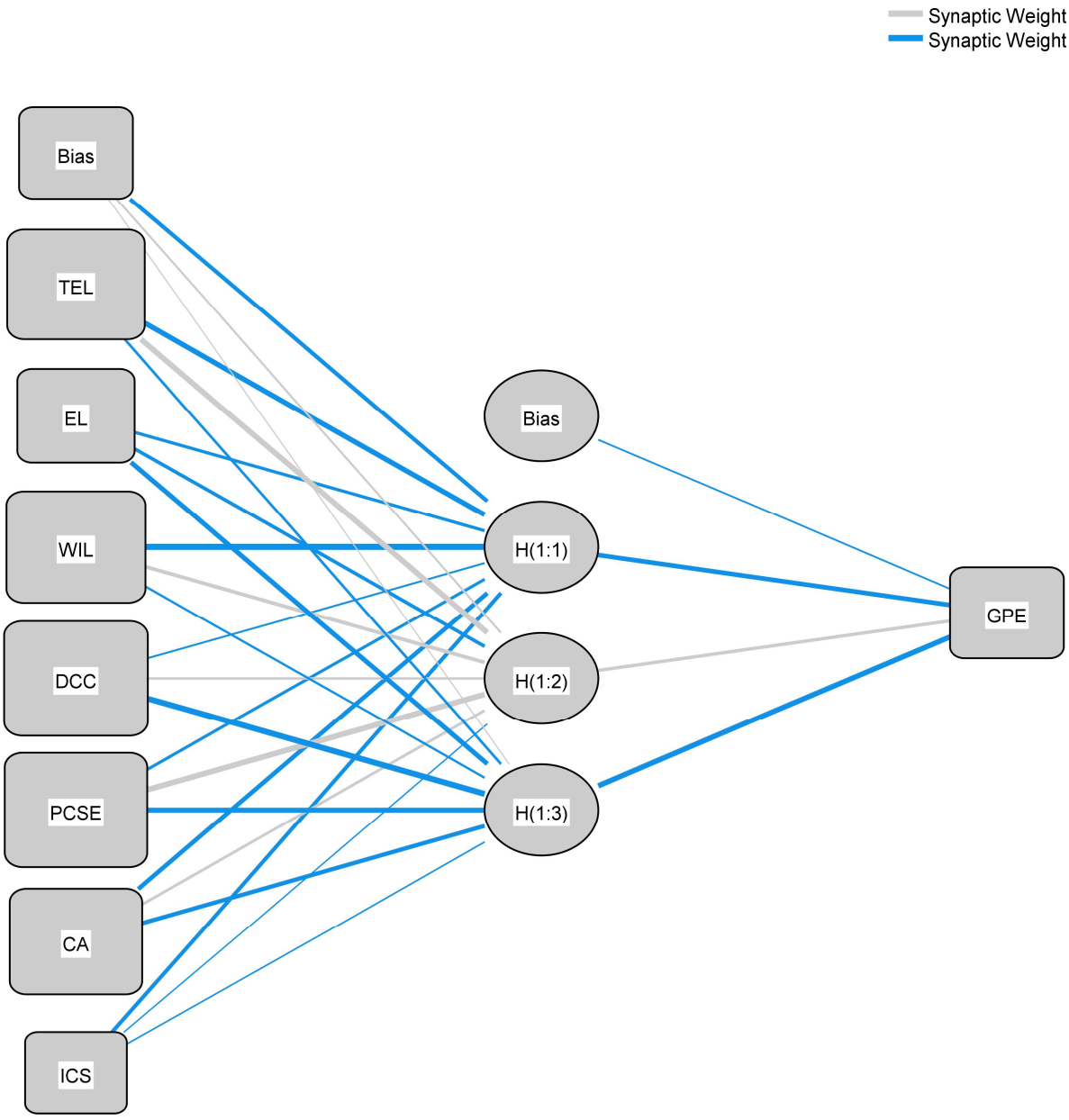
Case Processing Summary

		N	Percent
Sample	Training	2167	69.9%
	Testing	933	30.1%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

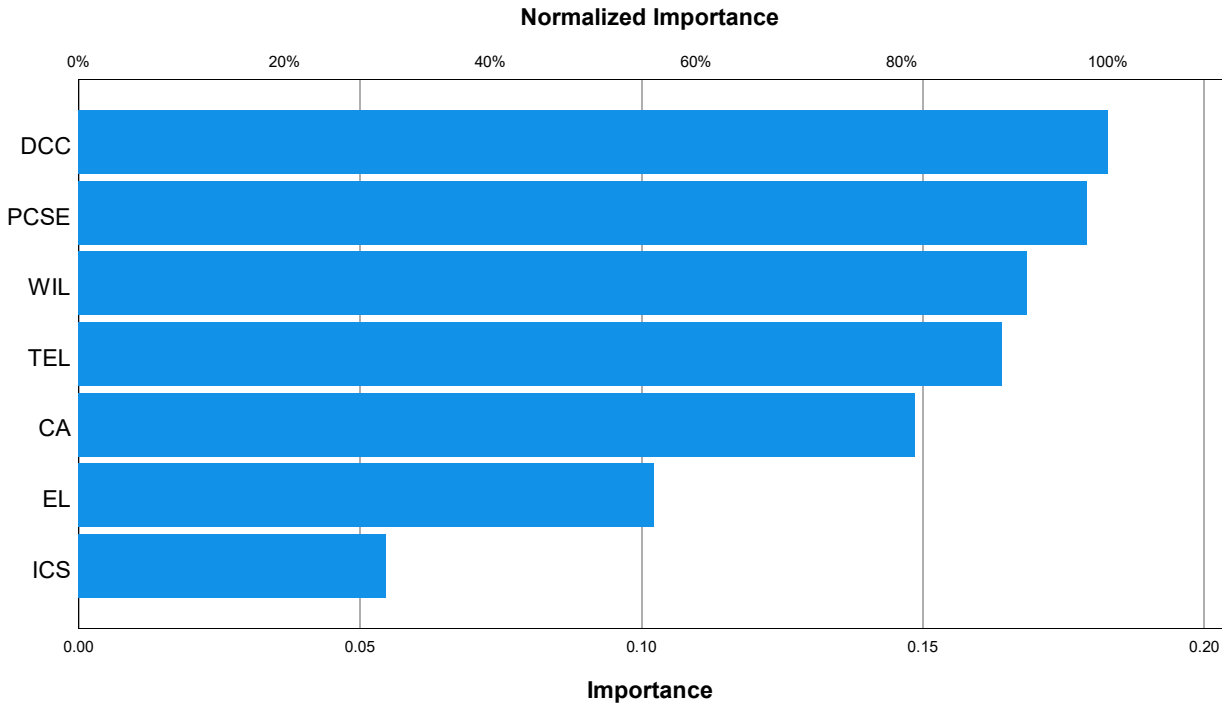
Training	Sum of Squares Error	755.107
	Relative Error	.697
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.12
Testing	Sum of Squares Error	329.279
	Relative Error	.710

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.164	89.7%
EL	.102	55.9%
WIL	.168	92.1%
DCC	.183	100.0%
PCSE	.179	98.0%
CA	.149	81.2%
ICS	.055	29.9%



Multilayer Perceptron

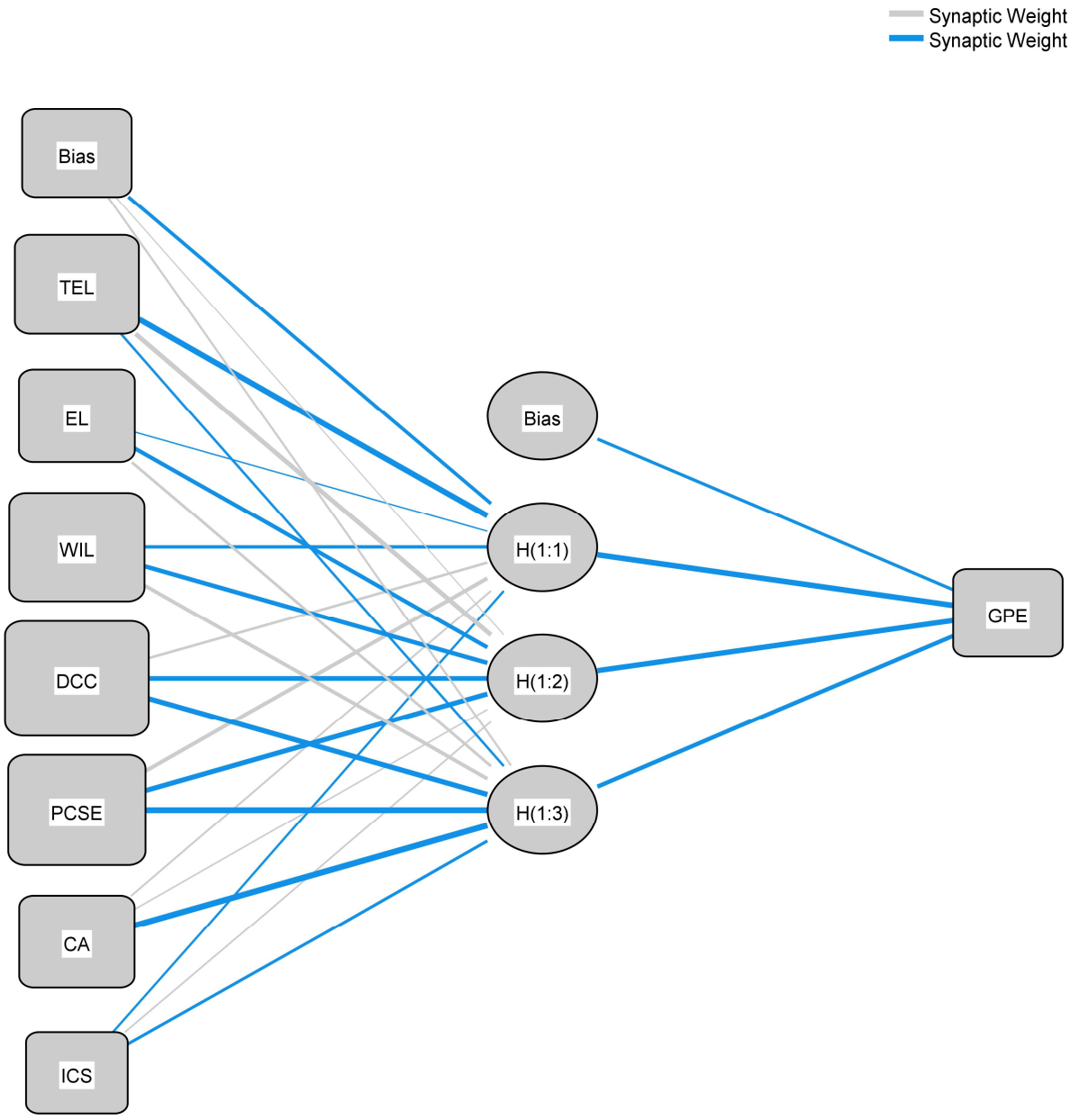
Case Processing Summary

		N	Percent
Sample	Training	2138	69.0%
	Testing	962	31.0%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

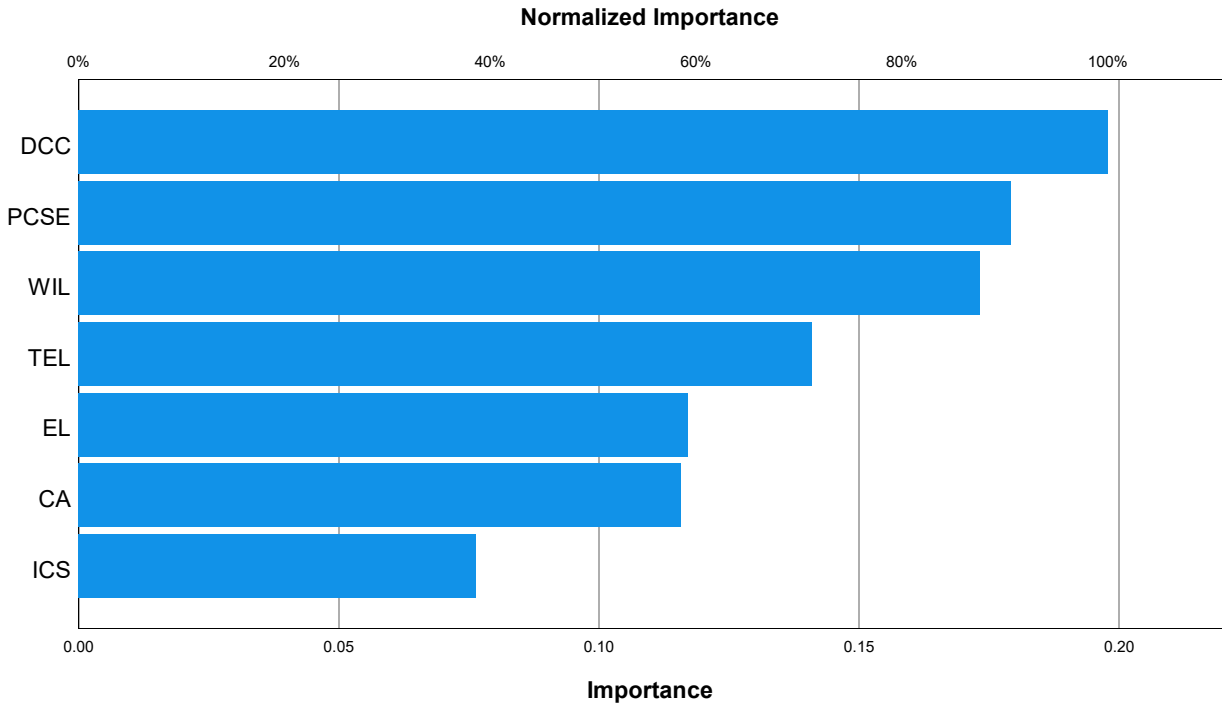
Training	Sum of Squares Error	754.818
	Relative Error	.706
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.11
Testing	Sum of Squares Error	332.360
	Relative Error	.711

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.141	71.2%
EL	.117	59.2%
WIL	.173	87.6%
DCC	.198	100.0%
PCSE	.179	90.5%
CA	.116	58.5%
ICS	.076	38.6%



Multilayer Perceptron

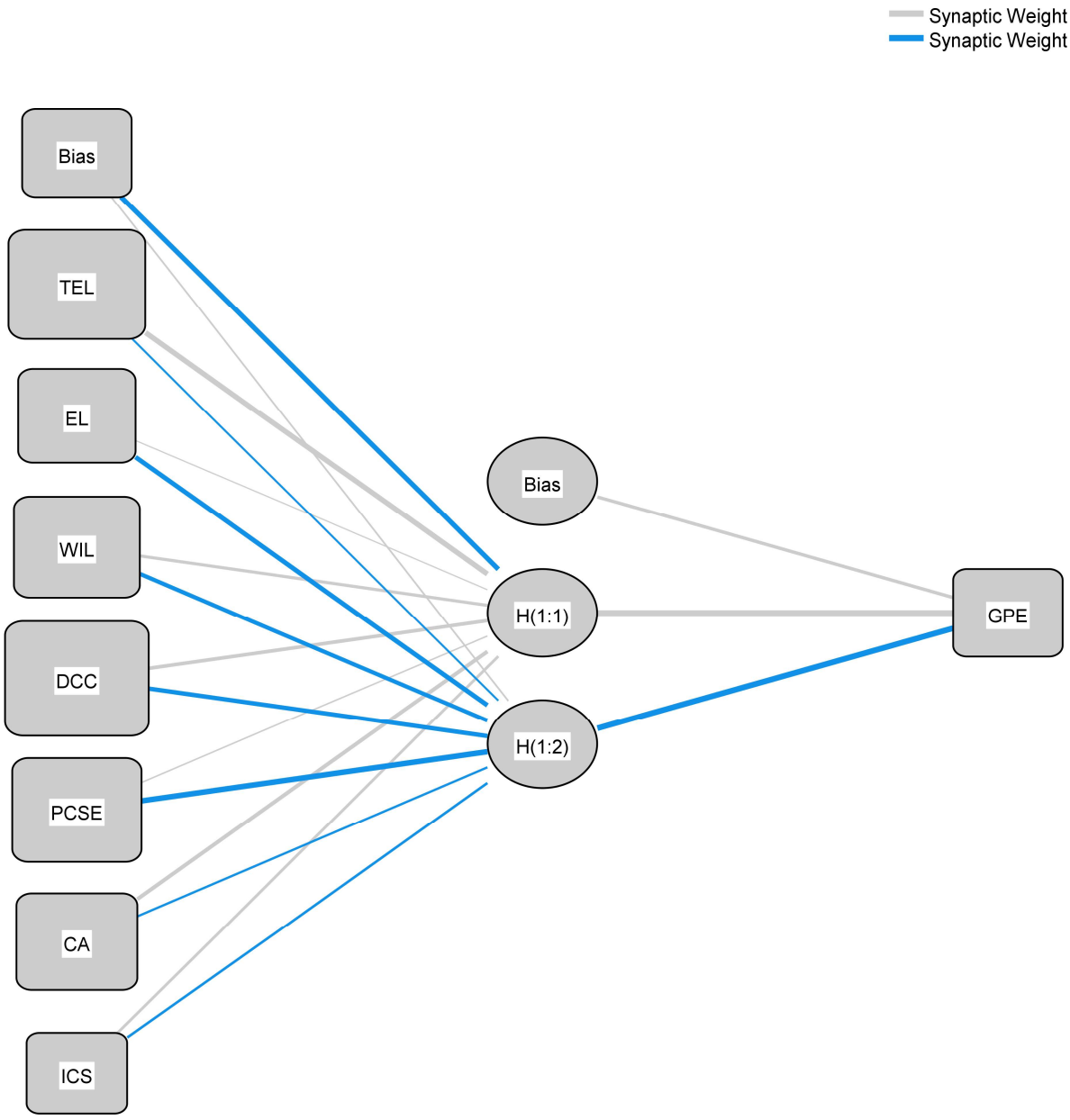
Case Processing Summary

		N	Percent
Sample	Training	2154	69.5%
	Testing	946	30.5%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		2
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

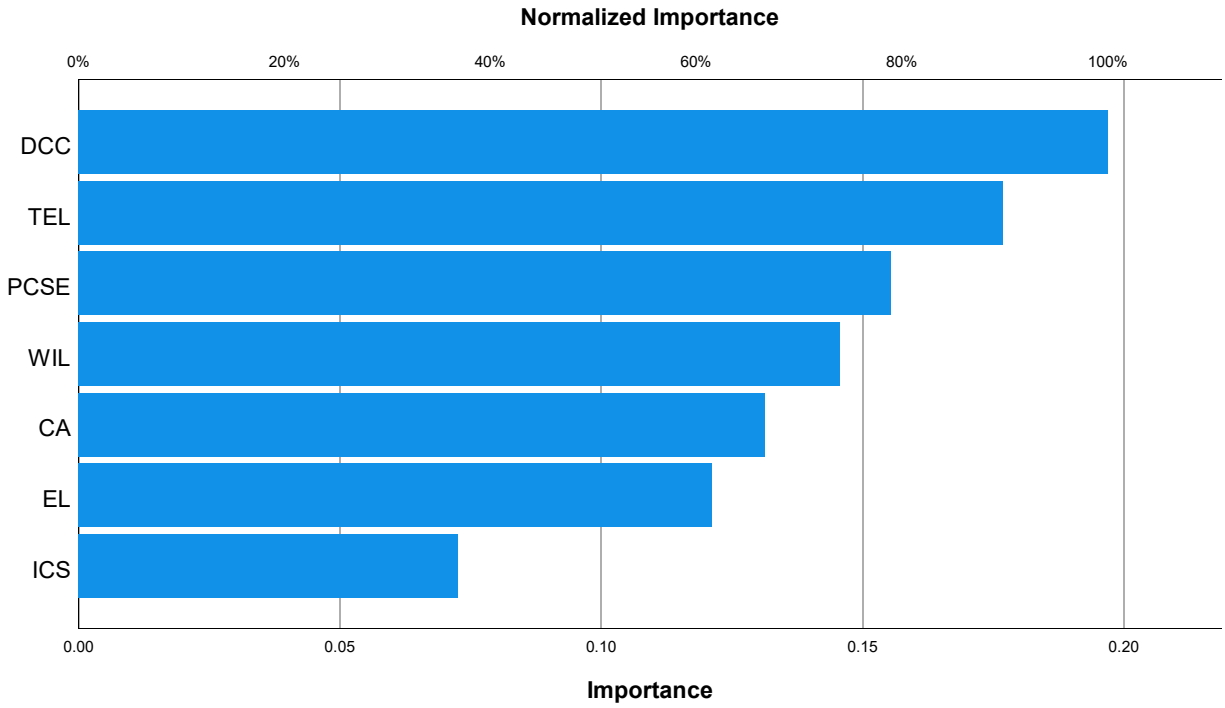
Training	Sum of Squares Error	752.319
	Relative Error	.699
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.10
Testing	Sum of Squares Error	318.605
	Relative Error	.681

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.177	89.8%
EL	.121	61.5%
WIL	.146	74.0%
DCC	.197	100.0%
PCSE	.155	78.9%
CA	.131	66.6%
ICS	.073	36.9%



Multilayer Perceptron

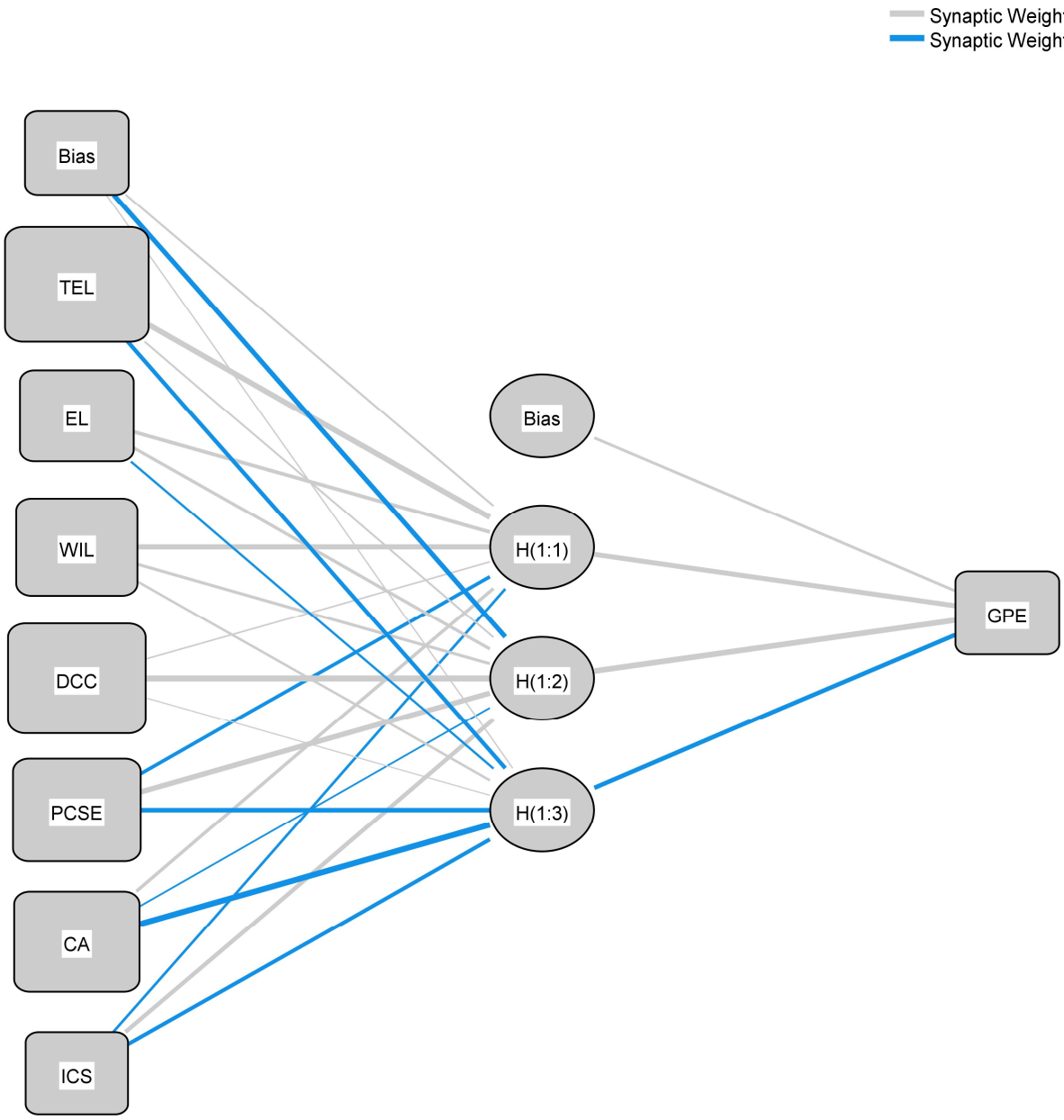
Case Processing Summary

		N	Percent
Sample	Training	2193	70.7%
	Testing	907	29.3%
Valid		3100	100.0%
Excluded		0	
Total		3100	

Network Information

Input Layer	Covariates	1	TEL
		2	EL
		3	WIL
		4	DCC
		5	PCSE
		6	CA
		7	ICS
	Number of Units[a]		7
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1[a]		3
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	GPE
	Number of Units		1
	Rescaling Method for Scale Dependents		Standardized
	Activation Function		Identity
	Error Function		Sum of Squares

a. Excluding the bias unit



Hidden layer activation function: Hyperbolic tangent

Output layer activation function: Identity

Model Summary

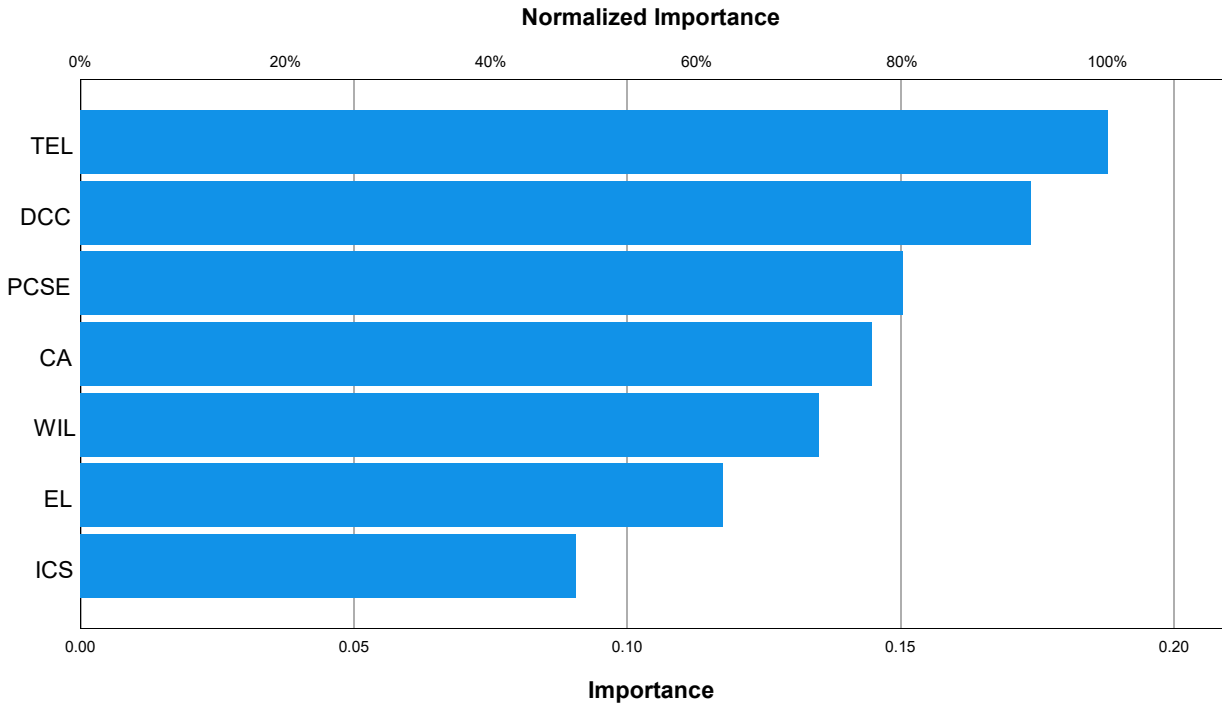
Training	Sum of Squares Error	774.709
	Relative Error	.707
	Stopping Rule Used	1 consecutive step (s) with no decrease in error[a]
	Training Time	0:00:00.10
Testing	Sum of Squares Error	328.035
	Relative Error	.672

Dependent Variable: GPE

a. Error computations are based on the testing sample.

Independent Variable Importance

	Importance	Normalized Importance
TEL	.188	100.0%
EL	.118	62.5%
WIL	.135	71.9%
DCC	.174	92.4%
PCSE	.150	80.0%
CA	.145	77.1%
ICS	.091	48.2%


Normalized Importance
0%
20%
40%
60%
80%
100%
TEL
DCC
PCSE
CA
WIL
EL
ICS
0.00
0.05
0.10
0.15
0.20
Importance